Final Year Design Project Presentation

Protein Sequence–Structure–Function Analytics and Predictive Modeling Using Deep Learning

Good morning, everyone. Today, I would like to present our Final Year Design Project, titled “Protein Sequence–Structure–Function Analytics and Predictive Modeling Using Deep Learning.” Our project belongs to the fields of Data Science, Artificial Intelligence, Machine Learning, Bioinformatics, and Computational Biology.

To understand our project, we first need to understand what a protein is. A protein is a biological molecule that performs many important jobs inside living organisms. Some proteins help digest food, some carry oxygen in the blood, some fight infections, and some help cells communicate with each other. In simple words, proteins are tiny biological workers inside our body.

Every protein is made from smaller building blocks called amino acids. These amino acids are connected in a specific order, forming a long chain. This order is called the protein sequence. We can think of it like beads arranged on a string. The sequence tells us which amino acid comes first, which comes second, and so on.

However, a protein does not remain as a straight chain. It folds into a specific three-dimensional shape. This folded shape is called the protein structure. The structure is very important because it affects what the protein can do. Two amino acids may be far apart in the sequence, but after folding, they may become close to each other in three-dimensional space.

Finally, every protein performs one or more biological jobs. These jobs are called protein functions. For example, a protein may bind to DNA, bind to calcium ions, help in a chemical reaction, or interact with another protein. Our project focuses on predicting these functions using artificial intelligence.

Our main research question is: Can we use deep learning to predict the molecular functions of a protein from its sequence and structure-related information?

To answer this question, we will use a publicly available research dataset called the DeepFRI PDB–Gene Ontology dataset. This dataset was released by the authors of a research paper titled “Structure-based Protein Function Prediction Using Graph Convolutional Networks,” published in Nature Communications in 2021.

The data in this dataset originally comes from the Protein Data Bank, also known as PDB. PDB is a large public library of experimentally determined protein structures. Scientists around the world determine protein structures in laboratories and upload them to PDB so that other researchers can use them.

The DeepFRI researchers selected protein chains from these experimental PDB structures. For each protein chain, they collected its amino-acid sequence, structural information, and known functional labels. They then processed this information into a format suitable for deep learning.

To connect PDB structures with functional information, they used a resource called SIFTS. SIFTS acts like a matchmaker between two databases. It helps identify which PDB protein structure belongs to which UniProt protein record. UniProt is another important biological database that provides protein sequences and functional information.

After connecting PDB and UniProt data, the DeepFRI researchers obtained functional annotations using Gene Ontology, commonly known as GO. Gene Ontology provides standardized labels for protein functions. Instead of writing functions in free text, it uses fixed labels such as “calcium ion binding,” “DNA binding,” or “protein binding.”

One protein can have more than one function. For example, a protein may bind calcium and also bind DNA. Therefore, our prediction task is called multi-label classification. This means that, for one protein, our model may predict several possible functions.

The dataset also contains structural information in the form of contact maps. A contact map is a simple way to represent which amino acids are close to each other in the protein’s three-dimensional structure. Instead of storing the entire complicated 3D structure, the dataset records which amino acids are near each other after the protein folds.

This is useful because the shape of a protein often influences its function. A contact map helps a machine-learning model understand important structural relationships without processing the full 3D coordinate file.

Our project will begin with dataset analysis. We will examine the number of proteins, sequence lengths, amino-acid composition, and the distribution of functional labels. We will also check which functions are common, which are rare, and how often different functions appear together. This statistical analysis will help us understand the dataset before training any model.

After that, we will develop deep-learning models. Our first model will be a sequence-based baseline model. It will use only the amino-acid sequence to predict protein functions. This model will help us understand how well sequence information alone can predict function.

Then, we will develop a second model that uses both sequence and structural information. In this model, the protein structure will be represented as a graph. Each amino acid will be treated as a node, and amino acids that are close in three-dimensional space will be connected by edges. A graph neural network will learn from these connections.

By comparing the sequence-only model with the sequence-and-structure model, we will investigate whether structural information improves function prediction. This comparison will form an important part of our research.

We will train our models using the training data and check their performance using validation data. Finally, we will evaluate the models on separate test data. Since this is a multi-label prediction problem, we will use evaluation metrics such as precision, recall, F1-score, F-max, and area under the precision-recall curve.

Along with the research work, we will also develop a software system. The system will be built using Python for data processing, model training, and prediction. We will use MySQL as our relational database to store protein information, functional annotations, prediction results, and experiment records. For the frontend, we plan to build a web interface where users can explore protein data and submit a protein sequence for prediction.

The user will enter or select a protein sequence. The backend will send this sequence to the trained deep-learning model. The model will return the most likely molecular functions along with prediction scores. The frontend will display these results in a clear and understandable way.

Our expected outcome is a complete research-based system that can analyze protein data and predict likely protein functions. We also expect to produce statistical insights about the dataset, trained machine-learning models, comparison results, and a working software prototype.

However, we also understand the limitations of this project. Our model will make computational predictions, not laboratory-verified biological facts. Some functions may be missing from the dataset, and some functions may have very few examples. Therefore, we will carefully analyze the data and clearly report the limitations of our results.

In the future, this type of system can be useful in biological research, medicine, and drug discovery. For example, if researchers discover a new protein sequence, a predictive model may help them understand what functions it might perform. This can save time and guide further laboratory experiments.

In conclusion, our project combines biology, data science, artificial intelligence, and software engineering. We will use protein sequence and structural data to train deep-learning models that predict molecular functions. We will also perform statistical analysis and build a complete software system using Python, MySQL, and a web-based frontend. Through this project, we hope to gain practical experience in machine learning, bioinformatics, database design, and full-stack software development, while contributing to an important area of computational biology.

Thank you.